

Linear systems and elimination

Exact extremizers for partial pivoting on orthogonal matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: open.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let L_n be the real unit lower triangular matrix whose entries strictly below the diagonal are all -1 . Define Q_n by the unique QR factorization $L_n = Q_n R_n$ with positive diagonal in R_n . For Q_n , use partial pivoting with the first available row chosen in a tie. Is it true that, for every $n \geq 2$,

$$\sup_{Q \in O(n)} \rho_{\text{PP}}(Q) = \rho_{\text{PP}}(Q_n), \quad O(n) = \{Q \in \mathbb{R}^{n \times n} : Q^T Q = I\}?$$

The supremum on the left includes all admissible partial-pivoting paths. The candidate is fully specified by L_n , rather than by an approximate numerical optimizer. This asks for the sharp extremizer, beyond the established exponential order of orthogonal growth.

Reference

Peca-Medlin, *Growth factors of orthogonal matrices and local behavior of Gaussian elimination with partial and complete pivoting*, published in SIAM J. Matrix Anal. Appl. (2024), §3.2 and Appendix B. The paper conjectures this equality and establishes $\rho_{\text{PP}}(Q_n) = 2^{n-1}(1 + o(1))/\sqrt{3}$.

Status check

Searches for *GEPP orthogonal conjecture 2026* and the exact paper title found no proof of the extremal equality. The 2026 butterfly paper still identifies orthogonal partial-pivoting growth as open. The August Shah–Urschel results concern different growth questions and pivot strategies; their exponential examples do not establish this exact supremum.